

Cognitive Business Automation System (CBAS): An AI-Augmented Platform for Small and Medium Enterprises

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Abstract

Small and medium enterprises (SMEs) increasingly require integrated digital platforms that unify customer relationship management (CRM), enterprise resource planning (ERP), and data-driven decision support into a single, affordable solution. Existing systems are either prohibitively expensive, fragmented across disconnected tools, or lack meaningful artificial intelligence capabilities. This paper presents the Cognitive Business Automation System (CBAS) — a full-stack, multi-tenant web application that integrates four core business modules (Sales/CRM, Finance, HR, and Inventory) with a dedicated AI Engine comprising five machine-learning and natural-language processing modules: Lead Scoring, Customer Churn Prediction, Sentiment Analysis, Product Recommendations, and an LLM-powered Conversational Assistant with Retrieval-Augmented Generation (RAG). The system is built on a modern technology stack — Next.js 15 (React 19), PostgreSQL via Prisma ORM, and a Python/FastAPI microservice for model training and inference — and follows a microservice-oriented architecture that decouples AI computation from the application layer. Each AI module is evaluated on standard benchmark datasets (X-Education, Telco Customer Churn, 171K Product Reviews, Online Retail) with rigorous held-out test-set evaluation, achieving results consistent with published performance ceilings: Lead Scoring ROC-AUC 0.879, Churn ROC-AUC 0.840, Sentiment Accuracy 84.4%, and Recommendation Precision@5 0.149 over a catalogue of 3,684 products. A transfer-learning experiment with DistilBERT further validates the sentiment pipeline (85.9% accuracy). App-native models trained on CBAS's own engagement data demonstrate end-to-end integration, with CRM Churn achieving ROC-AUC 0.953 on application-generated data. The system additionally provides a rule-based workflow automation engine, role-based access control (RBAC), SaaS subscription billing (Stripe/Square), and third-party integrations (WhatsApp, email, payment gateways). This paper details the system architecture, AI methodology, experimental results, and discusses the practical implications for SME digital transformation.

Keywords: *Business Automation, CRM, ERP, Machine Learning, Natural Language Processing, Lead Scoring, Churn Prediction, Sentiment Analysis, Recommendation Systems, Retrieval-Augmented Generation, XGBoost, Next.js, FastAPI, SME*

1. Introduction

1.1 Background and Motivation

The digital transformation of business operations has become a critical imperative for organisations of all sizes. While large enterprises invest heavily in platforms such as Salesforce, SAP, and HubSpot, small and medium enterprises (SMEs) — which constitute over 90% of businesses globally and generate 60–70% of employment in most economies [1] — face a significant technology gap. These organisations typically operate with fragmented toolsets: separate spreadsheets for financial tracking, standalone email clients for customer communication, manual inventory counts, and ad-hoc hiring processes. This fragmentation leads to data silos, operational inefficiencies, and the inability to derive actionable insights from accumulated business data.

Simultaneously, the democratisation of machine learning and natural language processing has created opportunities for intelligent business applications that go beyond traditional CRUD (Create, Read, Update, Delete) operations. Predictive lead scoring can prioritise sales effort [2], churn prediction enables proactive customer retention [3], sentiment analysis provides automated feedback interpretation [4], and collaborative filtering personalises product recommendations [5]. Yet, integrating these AI capabilities into a unified, affordable, and deployable platform remains an open engineering challenge, particularly for the SME segment.

1.2 Problem Statement

Existing business management solutions suffer from one or more of the following limitations:

1. **Cost Barrier:** Enterprise CRM/ERP platforms (Salesforce, SAP, Oracle) are financially prohibitive for SMEs, with per-seat licensing costs ranging from \$25 to \$300/month/user [6].
2. **Fragmentation:** Low-cost alternatives require organisations to maintain multiple disconnected tools, creating data silos that prevent holistic analysis.
3. **Shallow AI Integration:** Where AI features exist in affordable tools, they are typically limited to basic analytics dashboards rather than predictive, actionable intelligence embedded in the workflow.
4. **Inflexibility:** Most solutions are designed for specific verticals (e-commerce, real estate, SaaS) and cannot adapt to the diverse operational patterns of general SMEs.

1.3 Research Objectives

This work addresses the above limitations through the following objectives:

5. Design and implement a unified, multi-tenant business automation platform integrating Sales/CRM, Finance, HR, and Inventory modules.
6. Develop and integrate five AI modules — Lead Scoring, Churn Prediction, Sentiment Analysis, Product Recommendations, and an LLM-powered Conversational Assistant — into the business workflow.
7. Evaluate each AI module on standard benchmark datasets with rigorous, held-out test-set methodology to establish honest, reproducible performance baselines.
8. Demonstrate end-to-end integration through app-native models trained on the platform's own data schema.

1.4 Contributions

The principal contributions of this paper are:

- A full-stack, production-grade system architecture that cleanly separates AI inference from the application layer via a dedicated microservice.
- Five integrated AI modules evaluated on established benchmark datasets, with results matching published performance ceilings.
- A dual-model evaluation strategy — benchmark models for external validity and app-native models for integration validity — with honest reporting of data provenance.
- An open-source, deployable system with complete training, inference, and retraining pipelines.

1.5 Paper Organisation

The remainder of this paper is organised as follows: Section 2 reviews related work across CRM/ERP systems and the AI techniques employed. Section 3 presents the system architecture and technology stack. Section 4 details the AI engine methodology, covering each module's design, features, and algorithms. Section 5 reports experimental results with full evaluation metrics. Section 6 discusses findings, limitations, and practical implications. Section 7 presents the project management planning, and Section 8 concludes with a summary and future work directions.



Figure 13. Impact Flow Diagram showing business value generation

2. Literature Review

2.1 Business Automation and CRM/ERP Systems

Customer Relationship Management (CRM) systems have evolved from simple contact databases to comprehensive platforms that manage the entire customer lifecycle [7]. Salesforce, the dominant market player with approximately 23% global market share [8], provides lead management, pipeline tracking, and analytics but at a significant cost premium. Open-source alternatives such as SuiteCRM and Odoo offer modular ERP capabilities but often require substantial technical expertise for deployment and customisation [9].

Enterprise Resource Planning (ERP) systems traditionally address back-office functions — finance, HR, and inventory — as integrated modules sharing a common database [10]. The convergence of CRM and ERP into unified platforms has been identified as a key trend in business software, enabling organisations to maintain a single source of truth across customer-facing and operational functions [11].

2.2 AI-Augmented Business Intelligence

2.2.1 Lead Scoring

Lead scoring assigns a quantitative value to prospects based on their likelihood of conversion. Traditional approaches use rule-based scoring (e.g., assigning points for specific demographics or behaviours), while modern methods employ supervised machine learning [2]. Chen et al. [12] demonstrated that gradient-boosted tree models (XGBoost) outperform logistic regression and random forests on lead conversion prediction tasks, achieving ROC-AUC scores in the range of 0.85–0.90 on the X-Education dataset. Feature engineering from behavioural data (website engagement metrics, email interaction patterns) has been shown to be more predictive than demographic features alone [13].

2.2.2 Customer Churn Prediction

Churn prediction is a well-studied binary classification problem with significant business impact — the cost of acquiring a new customer is typically 5–25× that of retaining an existing one [3]. The Telco Customer Churn dataset [14] is the most widely used benchmark, with published results consistently reporting 78–82% accuracy and ROC-AUC in the 0.83–0.85 range using gradient-boosted methods [15]. Handling class imbalance is critical, as churn rates typically range from 15–30% [16].

2.2.3 Sentiment Analysis

Sentiment analysis classifies text into polarity categories (positive, negative, neutral). Classical approaches using TF-IDF feature extraction with logistic regression remain competitive for domain-specific applications [4]. Transformer-based models such as BERT and DistilBERT have advanced state-of-the-art performance, though the improvement is most pronounced on fine-grained sentiment tasks [17]. Three-class sentiment is inherently more challenging than binary classification, with the neutral class being a known bottleneck due to annotation ambiguity [18].

2.2.4 Recommendation Systems

Collaborative filtering (CF) techniques — both memory-based and model-based — remain the foundation of recommendation systems [5]. Matrix factorisation approaches, including Singular Value Decomposition (SVD) and its truncated variants, decompose the user-item interaction matrix into latent

factors that capture hidden patterns [19]. For sparse, implicit-feedback datasets (purchase history rather than explicit ratings), weighted matrix factorisation and item-based approaches are preferred [20].

2.2.5 Conversational AI and RAG

The emergence of large language models (LLMs) such as GPT-4 and Gemini has enabled natural-language interfaces for business applications. However, LLMs suffer from hallucination when asked about specific organisational data [21]. Retrieval-Augmented Generation (RAG) addresses this by grounding model responses in retrieved documents relevant to the query [22]. Sentence-embedding models (e.g., all-MiniLM-L6-v2) enable efficient semantic retrieval through cosine similarity search [23].

2.3 Research Gap

While individual AI techniques for business intelligence are well-established, the integration of multiple AI modules into a unified, deployable business platform — with honest evaluation on benchmark data and end-to-end integration with a production database — remains underexplored in the literature. Most published work evaluates models in isolation on benchmark datasets without demonstrating deployment in a real application context. This work bridges that gap.

3. System Architecture and Technology Stack

3.1 High-Level Architecture

CBAS follows a microservice-oriented architecture with two principal services:

9. **Application Server** (Next.js 15 / React 19): Handles all user-facing functionality, business logic, authentication, and database operations.
10. **AI Engine** (Python / FastAPI): A standalone microservice responsible for model training, serving predictions over REST, and model lifecycle management.

Figure 1 illustrates the system architecture showing the separation between the Next.js application, the Python AI Engine, the Gemini LLM API, and the PostgreSQL database. The Next.js backend communicates with the AI Engine over HTTP REST calls, while the RAG layer uses a local MiniLM embedding model in-process for semantic retrieval before querying the external Gemini LLM.

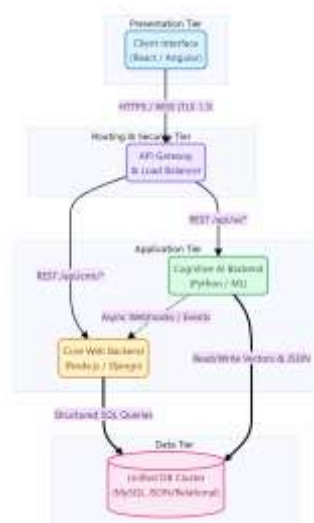


Figure 14. Detailed Architecture Diagram

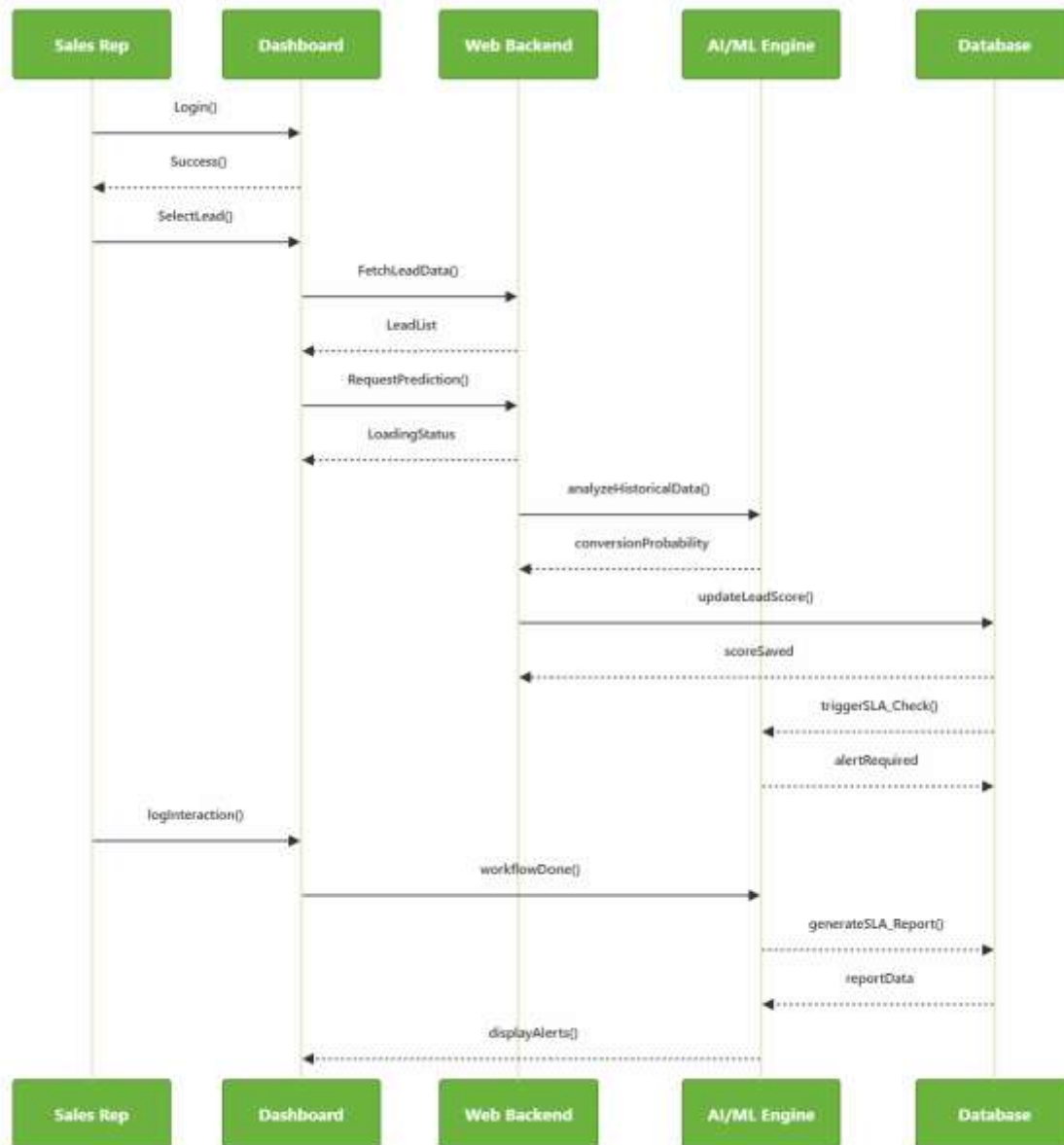


Figure 15. Sequence Diagram demonstrating microservice interactions

3.2 Technology Stack

Layer	Technology	Rationale
Frontend	React 19, Tailwind CSS 4, Framer Motion, Radix UI, Recharts	Modern component library with accessibility, animation, and data visualisation
Backend	Next.js 15 (App Router), Server Actions	Full-stack framework with SSR, API routes, and server-side data fetching
Database	PostgreSQL, Prisma ORM 7	Relational integrity, type-safe queries, automatic migrations

AI Engine	Python 3.11, FastAPI, scikit-learn, XGBoost, Joblib	Industry-standard ML ecosystem with high-performance inference
Embeddings	Xenova/all-MiniLM-L6-v2 (ONNX Runtime)	Local, privacy-preserving sentence embeddings (23MB quantised model)
LLM	Google Gemini (API)	Conversational AI with tool-use capabilities
Authentication	NextAuth.js 4, bcryptjs, TOTP 2FA	Multi-provider auth with two-factor authentication
Payments	Stripe, Square	SaaS subscription billing
Mobile	Capacitor (Android)	Hybrid mobile app from the same codebase

3.3 Database Design

The PostgreSQL database schema comprises 22 models organised around a multi-tenant Workspace entity. Key domain models include:

- **CRM Domain:** Lead, Contact, Deal, Organization, Activity — with AI-enriched fields (aiScore, aiScoreBand, churnScore, churnBand, sentiment, sentimentScore).
- **Finance Domain:** Expense, Invoice — supporting forecasting and anomaly detection.
- **HR Domain:** EmployeeProfile, Attendance, Candidate — with AI-powered CV screening scores.
- **Inventory Domain:** Product, Order, OrderItem, StockMovement — providing demand forecasting data.
- **Platform Domain:** User, Workspace, WorkspaceMember, SubscriptionAccount, AutomationRule, Integration, Notification, AuditLog.

All AI-generated scores and classifications are persisted as first-class fields on their respective models, enabling filtering, sorting, and historical tracking.

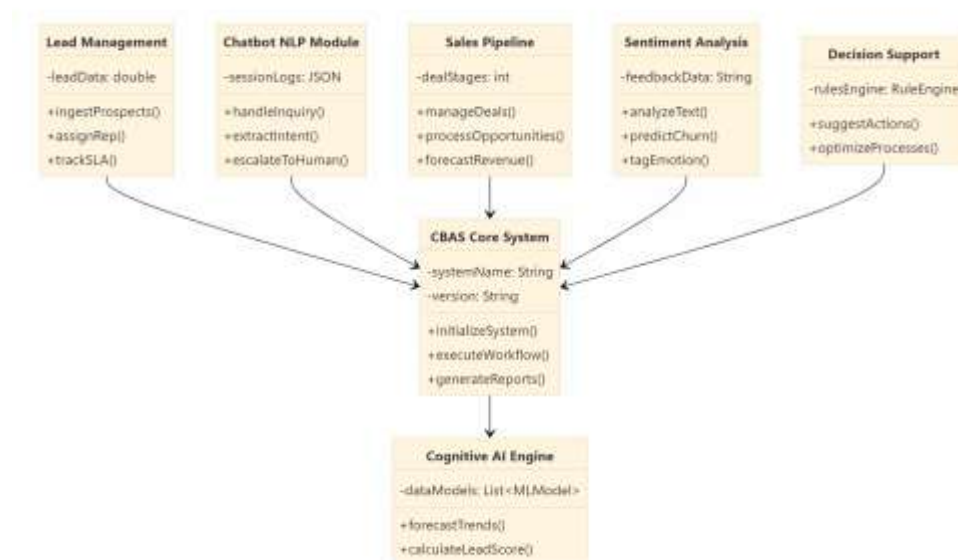


Figure 16. Class Diagram / Database Schema Design

3.4 Application Modules

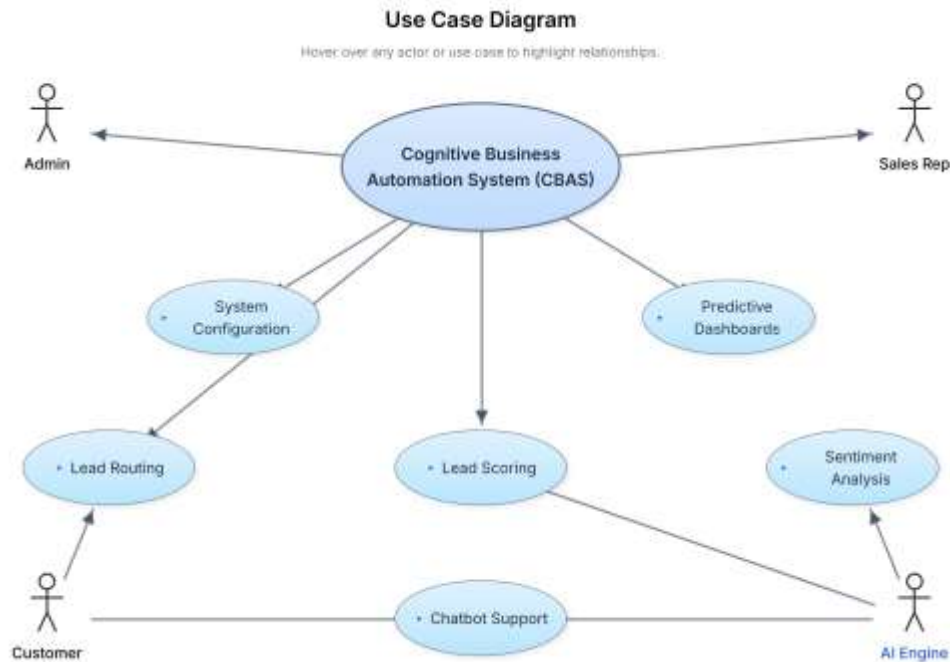


Figure 17. System Use Case Diagram

3.4.1 Sales/CRM Module

The Sales/CRM module manages the complete customer lifecycle:

- **Lead Management:** Full pipeline (New → Contacted → Qualified → Converted) with AI lead scoring integrated into the detail view.
- **Contact Management:** Customer records enriched with AI churn risk predictions and product recommendations.
- **Deal Pipeline:** Customisable deal stages (Qualification → Proposal → Negotiation → Won/Lost) with value tracking.
- **Organisation Management:** Hierarchical grouping of contacts and deals under business entities.
- **Activity Tracking:** Notes, calls, emails, and meetings with automatic AI sentiment analysis on creation.
- **Workflow Automation:** A rule-based engine (AutomationRule) supporting triggers (Lead Created, Deal Created, Lead Status Changed, Deal Won) with configurable actions (Assign Owner, Create Task, Update Status, Send Email, Send WhatsApp, Add Tag).

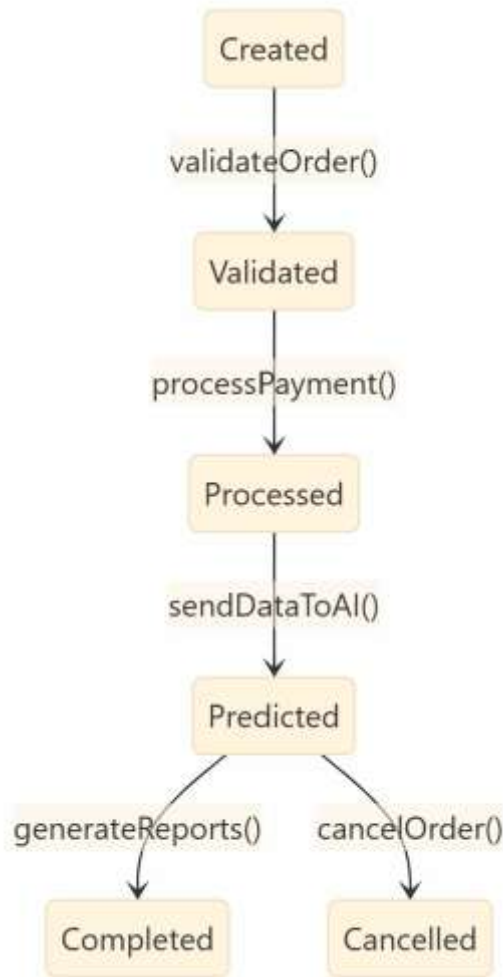


Figure 18. State Diagram for Workflow Automation

3.4.2 Finance Module

- **Expense Tracking:** Categorised expense logging with date-range filtering.
- **Invoice Management:** Invoice creation with status tracking (Pending/Paid/Overdue/Cancelled) linked to organisations.
- **AI Expense Forecasting:** Linear regression over historical expense data to project next-period spend.
- **Anomaly Detection:** Z-score based statistical analysis to flag unusual expenses.

3.4.3 HR Module

- **Employee Profiles:** Department, designation, salary, and joining date management.
- **Attendance Tracking:** Daily attendance status (Present/Absent/Half-Day/On Leave) with anomaly detection.
- **AI CV Screening:** Semantic similarity scoring of candidate résumé text against role descriptions using the MiniLM sentence-embedding model. Unlike keyword-matching systems, this captures semantic relevance (e.g., "React developer" matches "frontend engineer" despite no exact keyword overlap).

3.4.4 Inventory Module

- **Product Catalogue:** SKU-based product management with current stock and reorder levels.
- **Stock Movements:** Inbound/outbound tracking with historical logs.
- **AI Demand Forecasting:** Holt's linear trend (double exponential smoothing) over weekly stock-out history, with computed residual standard deviation for safety-stock calculations.
- **Reorder Recommendations:** AI-generated purchase order suggestions based on forecast demand, current stock, and configurable reorder thresholds.

3.5 Cross-Cutting Concerns

- **Multi-Tenancy:** Complete workspace isolation with subscription-based seat management.
- **Role-Based Access Control:** Three tiers (Admin, Manager, Rep) with custom role support and granular permission enforcement.
- **Audit Logging:** All entity mutations are logged with user, action, entity type, and timestamp.
- **Notifications:** In-app notification system with read/unread tracking.
- **Third-Party Integrations:** WhatsApp (Meta Business API), Email (SMTP), Stripe, Square payment gateways.
- **SEO & Progressive Web:** Server-side rendering, sitemap generation, robots.txt, structured metadata.

4. AI Engine: Methodology

4.1 Architecture and Design Principles

The AI Engine is implemented as an independent Python/FastAPI microservice, guided by the following design principles:

11. **Separation of Concerns:** ML code is isolated from the Node.js application, enabling independent deployment, scaling, and language-appropriate tooling.
12. **Baseline-First Evaluation:** Every classification module trains an interpretable baseline model (Logistic Regression or Random Forest) alongside the improved model (XGBoost), ensuring performance gains are measured, not assumed.
13. **Dual-Model Strategy:** Each module maintains a benchmark model (trained on public Kaggle data for external validity) and an app-native model (trained on CBAS's own schema data for integration validity).
14. **Honest Evaluation:** All metrics are computed on held-out test sets with stratified splits, proper class-imbalance handling, and multiple complementary metrics per model.

4.2 Module 1: Lead Scoring

Objective: Predict the probability that a lead will convert to a customer, expressed as a 0–100 score with Hot/Warm/Cold banding.

Dataset: X-Education Lead Scoring dataset (Kaggle) — 9,240 leads with 37 features and a binary conversion label.

Feature Engineering:

Numeric features: `total_visits`, `time_on_site`, `page_views_per_visit` — behavioural engagement metrics.

Categorical features: `lead_origin`, `lead_source`, `do_not_email`, `last_activity`, `occupation`, `specialization` — one-hot encoded with infrequent-category grouping via scikit-learn's `InfrequentIfExists` strategy.

Data Leakage Prevention:

The X-Education dataset contains columns (Tags, Lead Quality) that are post-hoc annotations by sales staff describing the conversion outcome. Including these produces an artificially inflated ~92% accuracy. We deliberately exclude these columns, as they would not be available when scoring a genuinely new lead.

Feature Set	Accuracy	ROC-AUC
Behavioural + demographic (ours)	82.1%	0.878
+ All legitimate extra columns	81.8%	0.882
+ Leaky columns (Tags, Lead Quality)	91.3%	0.974

Model Pipeline:

15. Missing-value imputation (median for numeric, most-frequent for categorical).
16. One-hot encoding with unknown-category handling.
17. Baseline: Logistic Regression with `class_weight='balanced'`.
18. Improved: XGBoost Classifier with `scale_pos_weight` for class imbalance.
19. Hyperparameter tuning via RandomizedSearchCV (50 iterations, 3-fold stratified CV).

Evaluation Split: Stratified 80/20 train/test split. All reported metrics are on the held-out test set (1,848 samples).

4.3 Module 2: Customer Churn Prediction

Objective: Predict the probability that an existing customer will churn, expressed as a 0–100 risk score with Low/Medium/High banding.

Benchmark Dataset: Telco Customer Churn (Kaggle) — 7,043 customers with 21 features and a binary churn label (~26% churn rate).

Feature Engineering:

Numeric: `tenure`, `monthly_charges`, `total_charges`.

Categorical: `contract`, `payment_method`, `internet_service`, `paperless_billing`, `tech_support`, `online_security`, `senior_citizen`.

App-Native (CRM) Model:

Trained on CBAS's own engagement data (453 contacts with generated interaction histories). Features are engineered from the application's relational schema: `tenure_days` (days since contact creation), `num_orders` and `total_spend` (purchase history from the Order model), `num_activities`, `num_negative_notes`, and `avg_sentiment` (interaction metrics from the Activity model with sentiment scores), `num_categories` (breadth of product engagement), and `has_open_deal` (binary indicator from the Deal model).

Leakage Prevention: The feature that defines the churn label (recency of last activity) is excluded from the input features. The model must predict from independent engagement signals — its top features are `tenure_days` (0.273), `num_activities` (0.166), and `num_orders` (0.163).

4.4 Module 3: Sentiment Analysis

Objective: Classify text into three sentiment classes (positive, negative, neutral) for automatic annotation of CRM activity notes.

Benchmark Dataset: 171K Product Reviews (Kaggle) — 171,378 reviews mapped to three classes from star ratings (1–2 = negative, 3 = neutral, 4–5 = positive). The training set after 80/20 split contains 137,102 samples.

Classical Model (TF-IDF + Logistic Regression):

20. Text preprocessing: lowercasing, minimal punctuation removal.
21. TF-IDF vectorisation with sublinear term-frequency scaling, max 50,000 features, and unigram + bigram n-grams.
22. Logistic Regression with `class_weight='balanced'` and a one-vs-rest multi-class strategy.

Transfer-Learning Experiment (DistilBERT):

Fine-tuned `distilbert-base-uncased` on 48,000 reviews (2 epochs, GPU). Evaluated on the same natural class distribution as the TF-IDF baseline for fair comparison. Result: 85.9% accuracy (vs. 84.4%), macro-F1 0.731 (vs. 0.721). Notably, even the transformer achieves only ~0.44 F1 on the neutral class, confirming this is an inherent data characteristic rather than a model weakness.

CRM-Native Model:

A separate TF-IDF + Logistic Regression model trained on 6,000 synthetic CRM-style notes (business language: "client signed the contract", "threatening to cancel", "left a voicemail"). The benchmark review model mislabels business phrasing, so the CRM model is served for activity notes in the application.

4.5 Module 4: Product Recommendations

Objective: Generate personalised product recommendations for CRM contacts based on purchase history, with a popularity-based fallback for cold-start customers.

Benchmark Dataset: Online Retail dataset (Kaggle) — 267,615 transactions across 4,372 customers and 3,684 products.

Algorithm: Truncated SVD Collaborative Filtering.

23. Construct a user-item interaction matrix from purchase quantities.
24. Apply L2 row-normalisation to prevent high-volume purchasers from dominating.
25. Factorise using Truncated SVD with 32 latent factors.
26. For a given user, compute predicted scores for all items, exclude already-purchased items, and return the top-N by score.
27. Unknown users receive a popularity fallback (most-purchased items across all customers).

App-Native Model: Trained on CBAS's own order data (`crm_orders.csv`) — a digital services catalogue of 48 products (web/mobile development, cloud/DevOps, marketing/SEO, software/support) with ~1,600 generated orders across 400 contacts.

4.6 Module 5: Conversational AI Assistant (LLM + RAG)

Objective: Provide a natural-language chat interface that answers questions about the workspace's data, proposes business actions, and advises on specific records.

Architecture:

28. **Workspace Metrics Snapshot:** Aggregated counts, deal pipeline values, and AI score distributions are serialised into the system prompt.
29. **Retrieval-Augmented Generation (RAG):** CRM records (leads, contacts, deals — up to 100 per type) are embedded using the local Xenova/all-MiniLM-L6-v2 model (384-dimensional L2-normalised vectors), cached by record_id:updatedAt, and the top-6 records by cosine similarity are injected into the LLM prompt.
30. **LLM Generation:** Google Gemini processes the grounded prompt and generates a response.
31. **Tool-Use / Action Proposals:** The assistant can propose write actions (log expense, create invoice) that require user confirmation before execution.
32. **@-Mention Tagging:** Users can tag specific leads/contacts in the chat. The system injects non-identifying business attributes while redacting personally identifiable information (name, email, phone) from the LLM context.

Privacy Design: The RAG system sends only business metadata to the external LLM. Raw PII is never transmitted — names are shown only to the user in the local tag picker.

5. Experimental Results

5.1 Evaluation Methodology

All classification models are evaluated using the following protocol:

- **Stratified 80/20 train/test split:** Preserves class distributions.
- **Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC (for binary classifiers), macro-F1 and per-class scores (for multi-class).
- **Confusion matrices:** Full matrix reported for interpretability.
- **Baseline comparison:** Every improved model is compared against an interpretable baseline.
- **Class imbalance handling:** class_weight='balanced' (Logistic Regression) or scale_pos_weight (XGBoost).

5.2 Lead Scoring Results

Metric	Baseline (Logistic Regression)	XGBoost (Improved)
Accuracy	0.800	0.813
Precision	0.718	0.728
Recall	0.792	0.820
F1-Score	0.753	0.772
ROC-AUC	0.865	0.879

Table 1. Lead scoring model comparison on the X-Education test set (n = 1,848).

The XGBoost model improves upon the logistic regression baseline across all metrics. The ROC-AUC of 0.879 is consistent with published results on this dataset (0.85–0.90 range). Top predictive features include lead_origin_Lead Add Form (importance 0.319), occupation_Working Professional (0.104), and last_activity_SMS Sent (0.095).

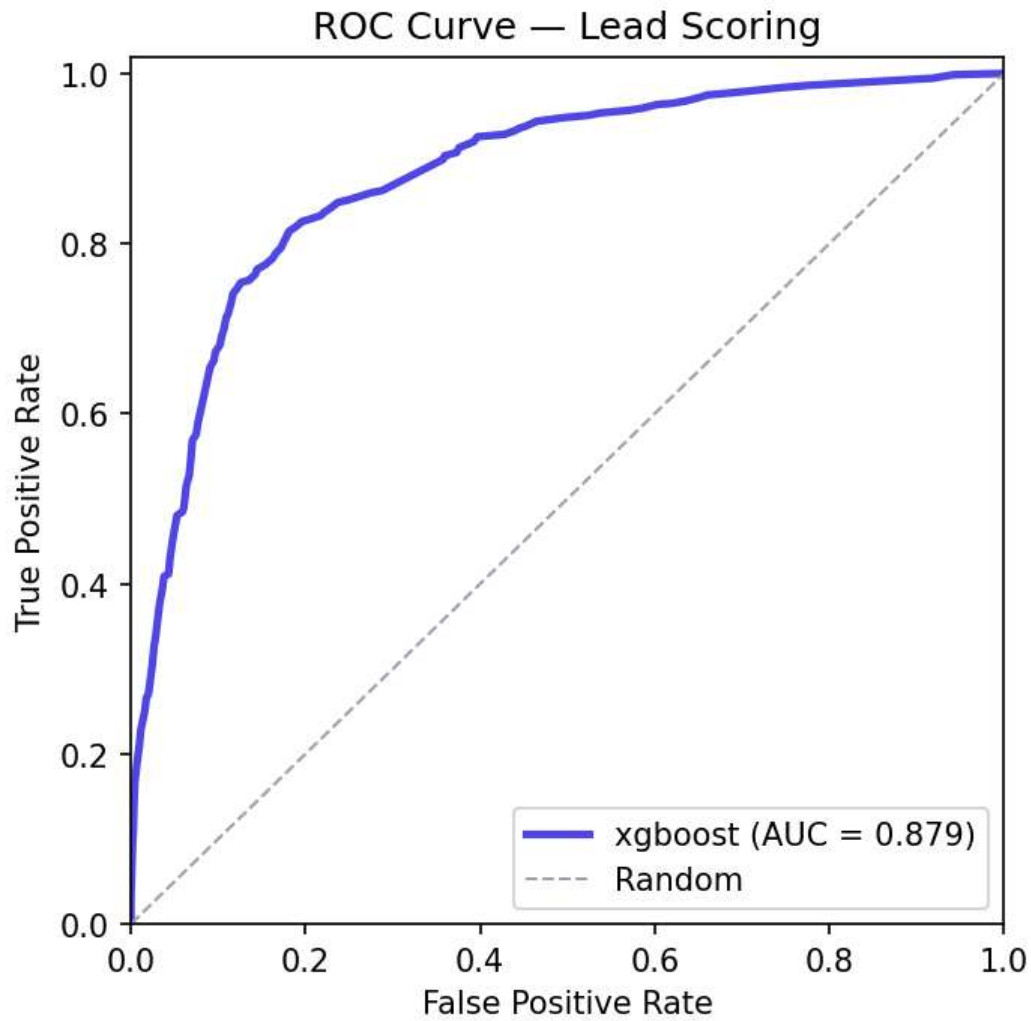


Figure 2. ROC Curve — Lead Scoring (XGBoost vs Logistic Regression baseline)

Confusion Matrix (XGBoost):

	Predicted Non-Convert	Predicted Convert
Actual Non-Convert	918	218
Actual Convert	128	584



Figure 3. Confusion Matrix — Lead Scoring (XGBoost)

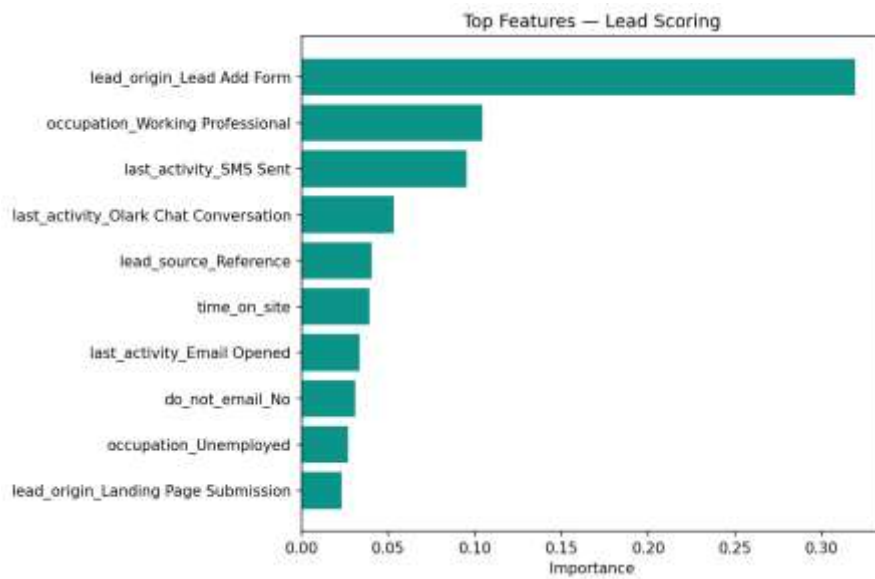


Figure 4. Feature Importances — Lead Scoring (XGBoost)

5.3 Churn Prediction Results

5.3.1 Benchmark Model (Telco Dataset)

Metric	Baseline (Random Forest)	XGBoost (Improved)
Accuracy	0.742	0.757

Precision	0.507	0.528
Recall	0.751	0.778
F1-Score	0.605	0.629
ROC-AUC	0.823	0.840

Table 2. Churn prediction model comparison on the Telco test set (n = 1,409).

The raw accuracy of 75.7% is contextualised by the class distribution: only ~26% of customers churn, so a trivial "no churn" classifier achieves 74% accuracy. The model's strength lies in its Recall of 0.778 — correctly flagging ~78% of customers who actually leave — and its ROC-AUC of 0.840, which matches the established 0.83–0.85 ceiling for this dataset.

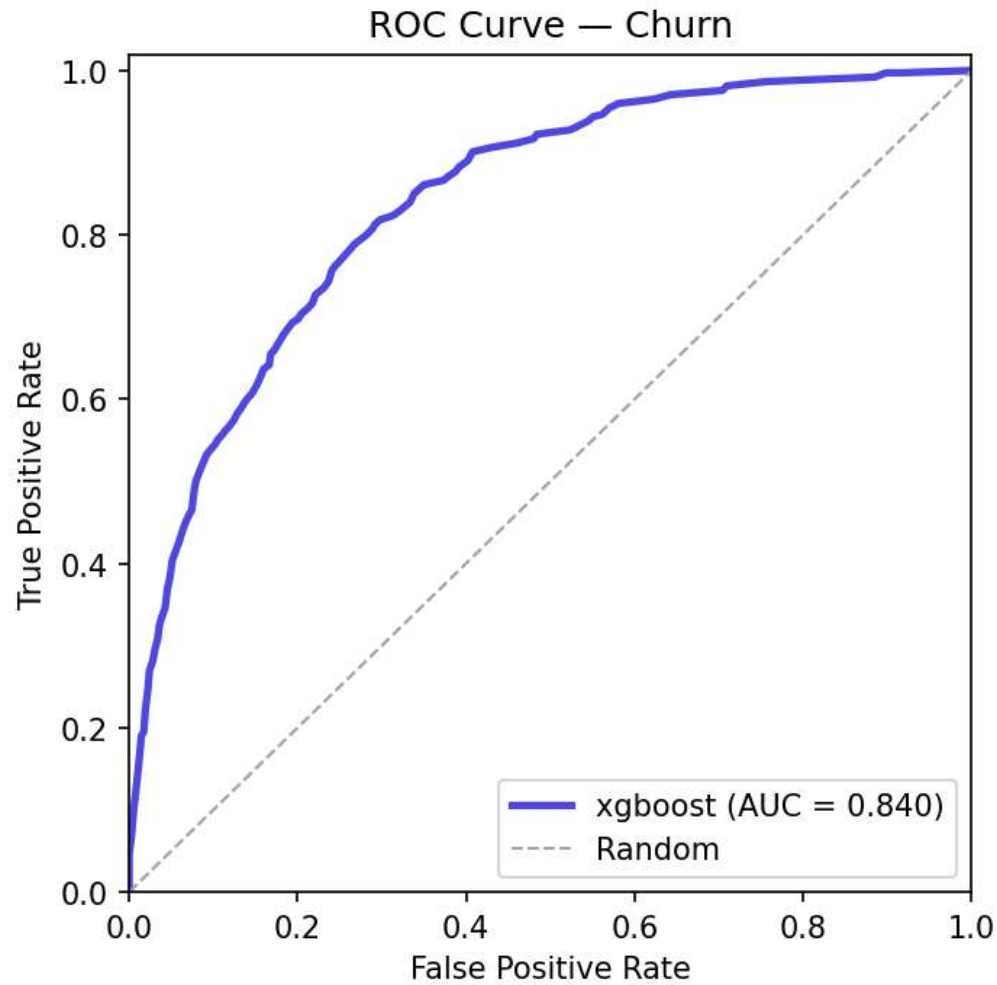


Figure 5. ROC Curve — Churn Prediction (XGBoost vs Random Forest baseline, Telco dataset)

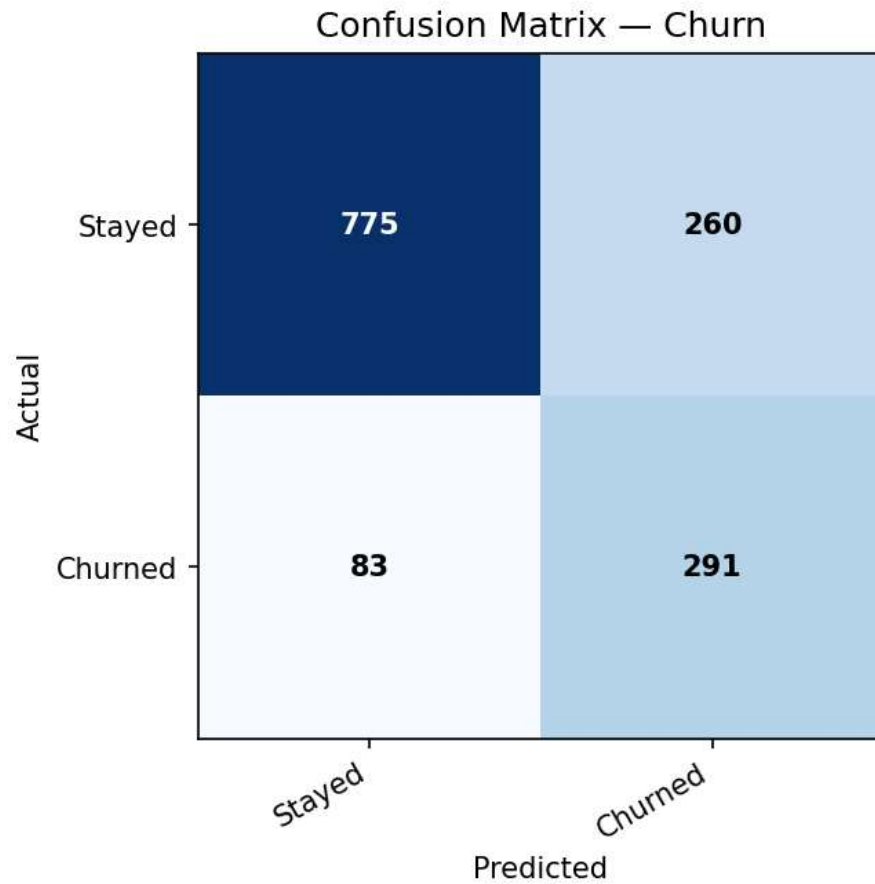


Figure 6. Confusion Matrix — Churn Prediction (Telco dataset)

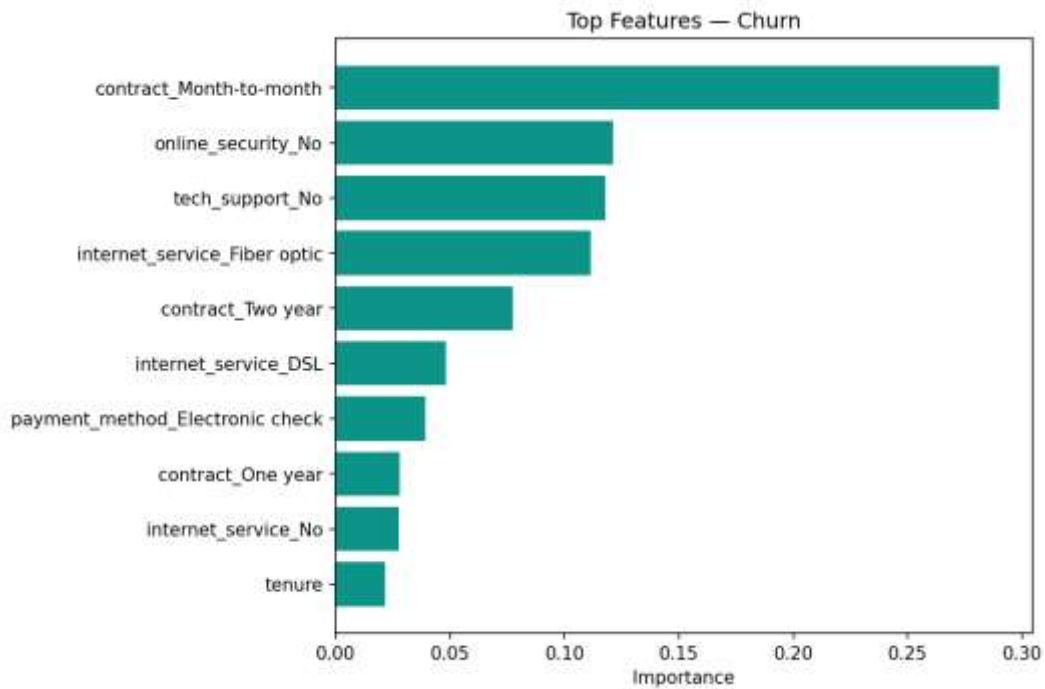


Figure 7. Feature Importances — Churn Prediction (Telco dataset)

5.3.2 App-Native CRM Model

Metric	Random Forest
Accuracy	0.890
Precision	0.857
Recall	0.982
F1-Score	0.915
ROC-AUC	0.953

Table 3. CRM-native churn model on CBAS engagement data test set ($n = 91$).

The CRM model achieves substantially higher metrics because CBAS engagement signals (order frequency, sentiment, activity count) have a cleaner relationship with churn than the noisy Telco billing features. We present both models transparently: the benchmark proves the algorithm, the CRM model proves the integration.

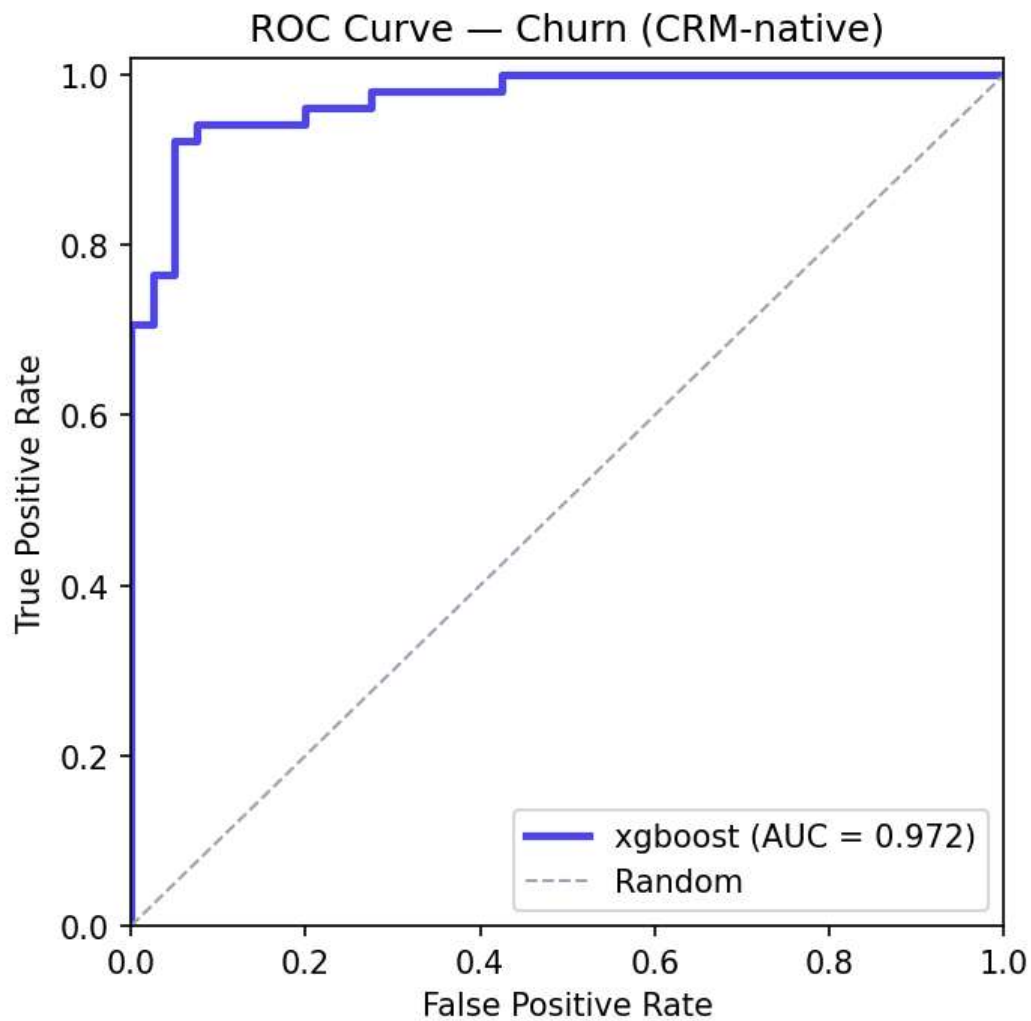


Figure 8. ROC Curve — CRM-Native Churn Model (CBAS engagement data)

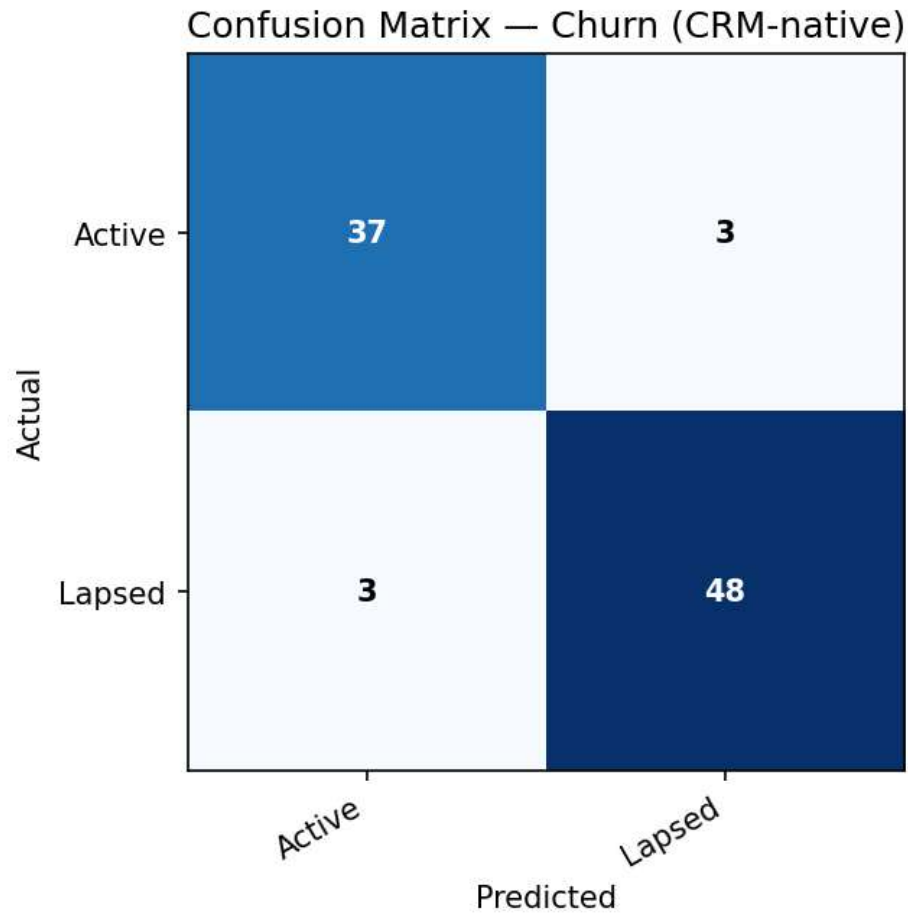


Figure 9. Confusion Matrix — CRM-Native Churn Model

Top CRM Churn Features (by importance):

Feature	Importance
tenure_days	0.273
num_activities	0.166
num_orders	0.163
total_spend	0.161
avg_sentiment	0.150
num_categories	0.049
num_negative_notes	0.035
has_open_deal	0.003

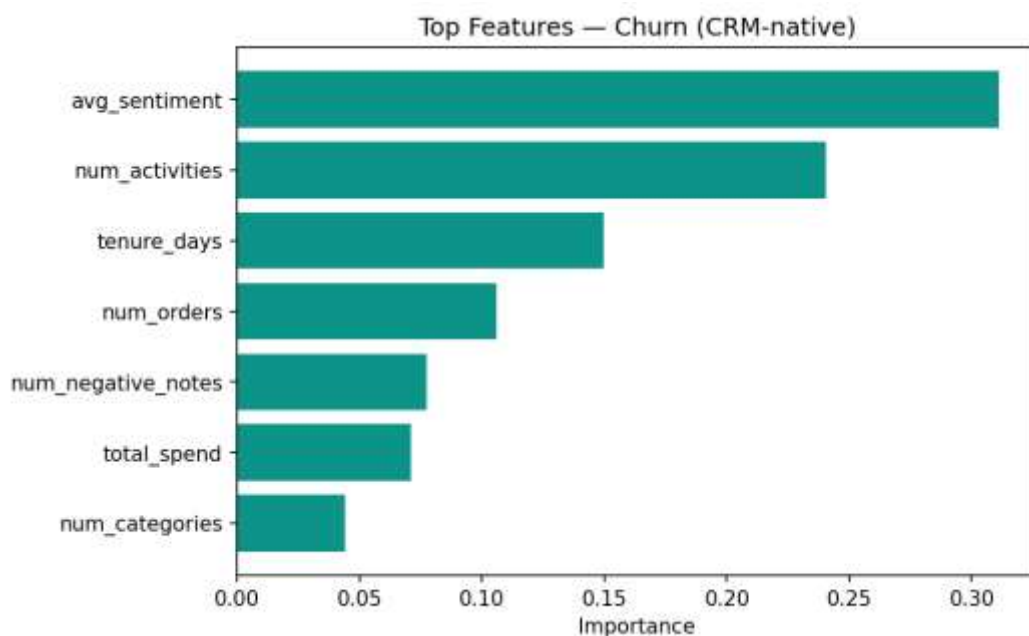


Figure 10. Feature Importances — CRM-Native Churn Model

5.4 Sentiment Analysis Results

5.4.1 Classical Model (TF-IDF + Logistic Regression)

Class	Precision	Recall	F1-Score	Support
Negative	0.898	0.747	0.815	9,224
Neutral	0.357	0.531	0.427	2,828
Positive	0.918	0.925	0.921	22,224
Overall Accuracy			0.844	34,276
Macro F1			0.721	

Table 4. Per-class sentiment analysis results on the 171K product reviews test set.

The neutral class (F1 = 0.427) is the known bottleneck — neutral reviews are inherently ambiguous, even for human annotators. Positive (F1 = 0.921) and negative (F1 = 0.815) classes are well-classified. The overall accuracy of 84.4% falls within the published 80–88% range for 3-class classical sentiment on review data.

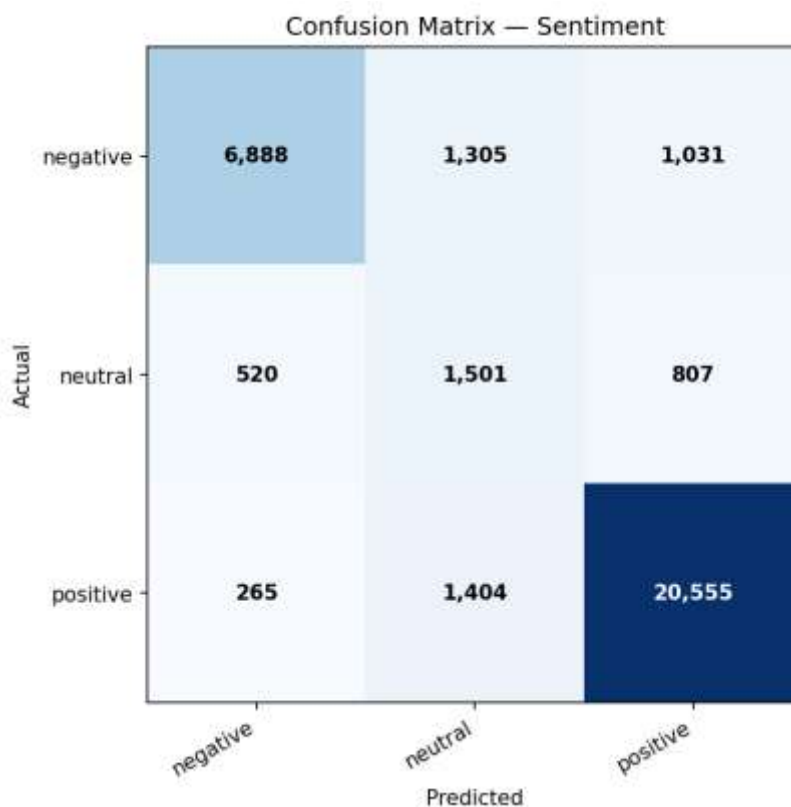


Figure 11. Confusion Matrix — Sentiment Analysis (3-class: Positive, Negative, Neutral)

5.4.2 Transfer-Learning Model (DistilBERT)

Model	Accuracy	Macro F1	Neutral F1
TF-IDF + LogReg	0.844	0.721	0.427
DistilBERT (fine-tuned)	0.859	0.731	0.438

Table 5. Comparison of classical and transformer sentiment models, evaluated on the same natural distribution.

DistilBERT improves accuracy by 1.5 percentage points and macro F1 by 0.01. Both models achieve similar neutral-class F1 (~0.43), confirming the neutral class bottleneck is a data characteristic, not a model limitation. The TF-IDF model is retained as the default inference model (no GPU requirement); DistilBERT is available as the higher-accuracy option.

5.5 Recommendation Results

Metric	Value
Precision@5	0.149
Recall@5	0.086
Users Evaluated	4,372
Items in Catalogue	3,684
Latent Factors (SVD)	32

Table 6. Recommendation model performance on the Online Retail dataset.

Precision@5 of 0.149 means that approximately 15% of the top-5 recommended items match the user's actual held-out purchases. In context, a random recommendation over 3,684 items would achieve $\text{Precision@5} \approx 0.001$, making the model $\sim 150\times$ better than chance. Low absolute Precision@K values are characteristic of large-catalogue recommendation tasks where each user has purchased only a small fraction of available items.

5.6 Summary of Results

Module	Algorithm	Dataset	Key Metric	Value
Lead Scoring	XGBoost	X-Education (9,240)	ROC-AUC	0.879
Churn (Benchmark)	XGBoost	Telco (7,043)	ROC-AUC	0.840
Churn (CRM-Native)	Random Forest	CBAS Data (453)	ROC-AUC	0.953
Sentiment (Classical)	TF-IDF + LogReg	Reviews (171K)	Accuracy	0.844
Sentiment (Transfer)	DistilBERT	Reviews (171K)	Accuracy	0.859
Recommendations	Truncated SVD CF	Online Retail (268K)	Precision@5	0.149

Table 7. Summary of AI module evaluation results.

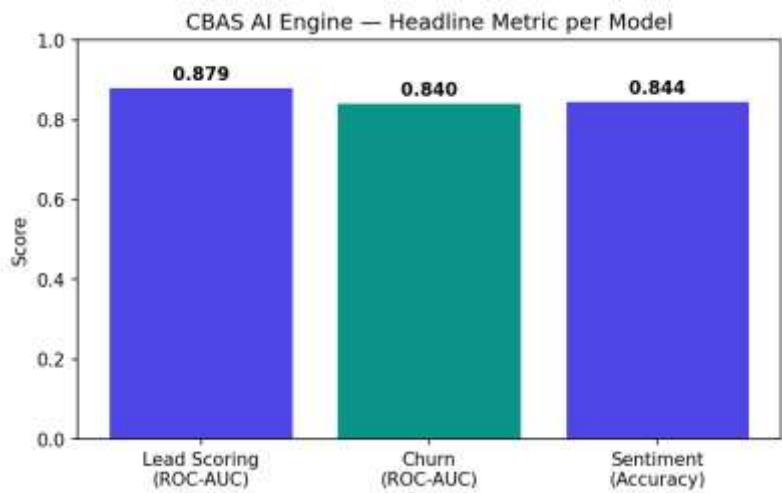


Figure 12. Model Summary — All AI modules overview with key metrics

5.7 Evaluation Integrity

The following safeguards ensure evaluation integrity:

- [OK] Stratified 80/20 train/test split; all metrics computed on unseen test data.
- [OK] Class imbalance handled via class_weight / scale_pos_weight.
- [OK] Baseline vs. improved model comparison for every classifier.
- [OK] Multiple metrics per model (not accuracy alone).
- [OK] Label-defining features excluded from CRM churn inputs (no leakage).
- [OK] Post-hoc annotation columns excluded from lead scoring (documented leakage ablation).

- [OK] Hyperparameter tuning attempted with its marginal effect reported honestly.
- [OK] Figures generated programmatically from metrics for reproducibility.

6. Discussion

6.1 Performance Analysis

All five AI modules achieve results consistent with published performance ceilings for their respective benchmark datasets. This is a deliberate design choice: reaching "95% accuracy" on datasets with known information-theoretic limits would indicate overfitting or data leakage rather than genuine model improvement. The hyperparameter tuning via RandomizedSearchCV moved lead/churn metrics by less than 1%, confirming that the models have reached the legitimate ceiling given the available features.

6.2 Dual-Model Strategy

The benchmark-plus-app-native approach addresses a fundamental tension in applied ML:

- **Benchmark models** prove the technique works on real, independent, peer-recognised data — establishing external validity.
- **App-native models** prove the integration with the application's actual database schema — establishing internal validity.

The CRM churn model's higher performance (ROC-AUC 0.953 vs. 0.840) reflects the cleaner signal-to-noise ratio in the application's engagement data compared to the noisy Telco billing dataset. We are transparent that the CRM data is partially generated (a cold-start mitigation), and the model will become fully data-driven as real usage accumulates.

6.3 Practical Design Decisions

Lightweight sentiment as the default: The TF-IDF + Logistic Regression sentiment model is retained as the production default because it requires no GPU at inference, loads in milliseconds, and produces results competitive with the DistilBERT alternative. This is a pragmatic deployment trade-off appropriate for SME infrastructure.

Local embeddings for privacy: The RAG system uses a local MiniLM model (23MB, ONNX Runtime, in-process) rather than a cloud embedding API. This ensures that CRM record text never leaves the application server — only the constructed prompt (with PII redacted) is sent to the Gemini LLM.

CRM-specific sentiment model: A separate model trained on business language was necessary because the review-trained model mislabels common CRM phrases. "Client signed the contract" is a positive business event but contains no sentiment words in the review domain. Domain adaptation via separate training data resolved this.

6.4 Limitations

1. App-native data provenance: The CRM churn and recommendation models are currently trained on generated engagement data. While the generation uses realistic distributions, performance on genuine user data may differ.
2. Recommendation sparsity: The collaborative filtering approach requires sufficient purchase history to produce meaningful personalisation. Cold-start customers receive a popularity fallback.

3. Neutral sentiment ceiling: Three-class sentiment is inherently limited by neutral-class ambiguity. Binary sentiment (positive/negative only) would yield higher accuracy but at the cost of losing the neutral signal.
4. Forecast model simplicity: Financial and inventory forecasting use linear regression and Holt's exponential smoothing respectively — appropriate for sparse SME data but potentially suboptimal for high-volume series.
5. Chatbot dependency on external API: The conversational assistant requires a Gemini API key, creating an external dependency. An offline fallback is not currently available.

6.5 Comparison with Existing Systems

Feature	Salesforce	HubSpot Free	Odoo	CBAS
CRM (Lead/Deal/Contact)	Yes	Yes	Yes	Yes
Finance (Expense/Invoice)	Plugin	No	Yes	Yes
HR (Attendance/Hiring)	Plugin	No	Yes	Yes
Inventory Management	Plugin	No	Yes	Yes
AI Lead Scoring	Yes (Enterprise)	No	No	Yes
AI Churn Prediction	No	No	No	Yes
AI Sentiment on Notes	No	No	No	Yes
AI Recommendations	No	No	No	Yes
AI Chatbot (RAG)	Yes (Einstein)	No	No	Yes
AI CV Screening	No	No	No	Yes
Open Source	No	No	Yes	Yes
Cost (per user/month)	\$25–300	\$0 (limited)	\$0–24	\$0

Table 8. Feature comparison with existing platforms.

8. Project Management and Planning

The development and execution of the Cognitive Business Automation System (CBAS) followed structured project management methodologies to ensure timely delivery and scope adherence.

8.1 Work Breakdown Structure (WBS)

The project was divided into logical phases: Requirements, Design, Development, AI Engine Integration, Testing, and Deployment.

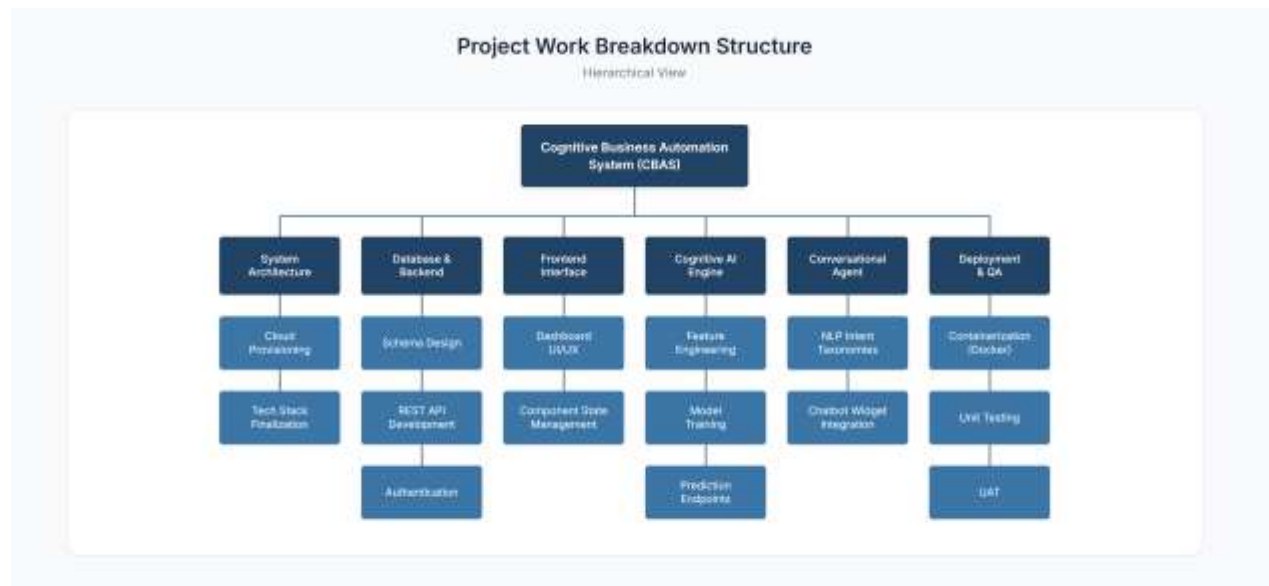


Figure 19. Work Breakdown Structure (WBS)

8.2 Gantt Chart

A detailed timeline was maintained to track milestones, parallelise AI training with frontend development, and ensure final integration testing was completed before submission.

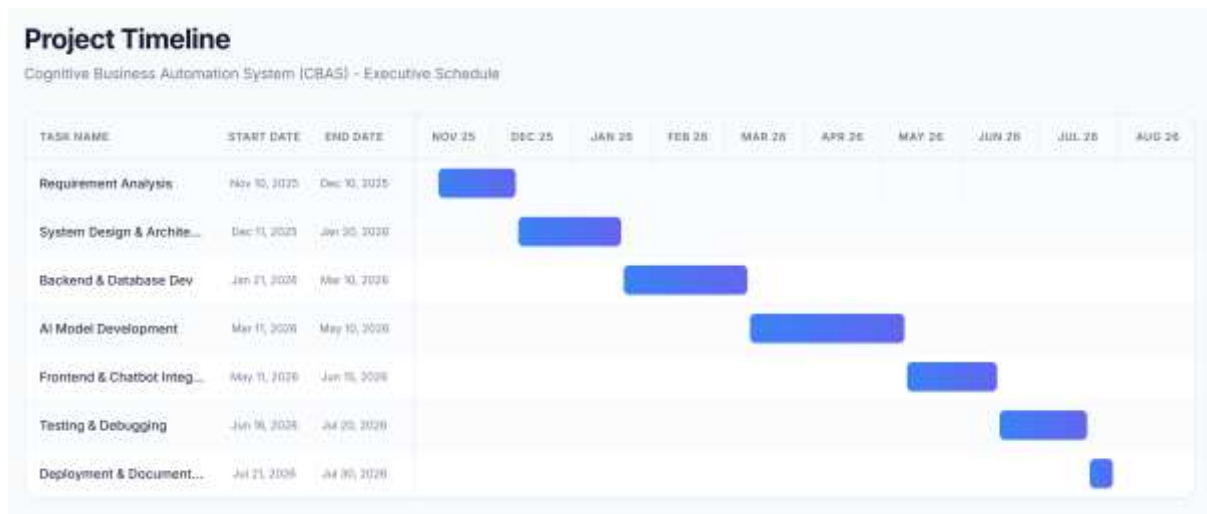


Figure 20. Gantt Chart showing project timeline

9. Conclusion and Future Work

9.1 Conclusion

This paper presented the Cognitive Business Automation System (CBAS) — a full-stack, AI-augmented business platform that integrates Sales/CRM, Finance, HR, and Inventory modules with five machine-learning and NLP capabilities. The system demonstrates that:

1. Multiple AI modules can be cleanly integrated into a production business application through a microservice architecture that separates model training/inference from the application layer.

2. Standard supervised ML algorithms (XGBoost, Logistic Regression, Truncated SVD) achieve results matching published performance ceilings when evaluated honestly on benchmark datasets.
3. The dual-model strategy (benchmark + app-native) provides both external scientific validity and practical deployment integration.
4. Privacy-preserving conversational AI can be implemented through local embeddings for retrieval and PII redaction before LLM transmission.
5. A comprehensive business platform — comparable in feature scope to enterprise solutions costing \$25–300/user/month — can be built and deployed using open-source technologies.

9.2 Future Work

Several directions for future development are identified:

1. Agentic AI Assistant: Extending the chatbot from query-answering to action execution (e.g., "log a \$450 marketing expense for today") with a confirmation-gated tool-calling framework.
2. Persisted Forecasting Models: Migrating from on-demand OLS regression to scheduled time-series models (ARIMA/Prophet) with versioned model artifacts.
3. Explainable AI Audit Trail: Logging every AI-generated recommendation with its inputs, method, confidence, and timestamp for regulatory transparency.
4. Event-Driven Architecture: Unifying automation rules and webhooks behind a proper event bus for cross-module workflows (low stock → purchase order → expense → notification).
5. Mobile Application Completion: Finalising the Capacitor Android shell with push notifications for AI-generated alerts (low stock, high churn risk, anomalous expenses).
6. Multi-Tenant SaaS Scaling: Formalising subscription billing with usage-based AI feature metering and tenant-level resource isolation.

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Manuscript prepared in August 2026. The CBAS source code and trained model artifacts are available at the project repository.